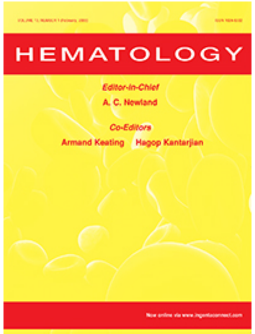

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Impact of pre-transplant use of antibiotics on the graft-versus-host disease in adult patients with hematological malignancies

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ABSTRACT

Objectives: Changes in fecal microbiota affect the incidence and extent of graft-versus-host disease (GVHD) after allogeneic hematopoietic stem cell transplantation (HSCT). Most patients with hematological malignancies receive antibiotics for the treatment of febrile neutropenia prior to allogeneic HSCT, and pre-transplant use of antibiotics may influence the fecal microbiota and GVHD.

Methods: We retrospectively analysed consecutive adult patients with hematological malignancies who received allogeneic HSCT at Chungnam National University Hospital between 2007 and 2018. Pre-transplant use of antibiotics was defined as the use of antibiotics before conditioning chemotherapy.

Results: This study included 131 patients with a median age of 46 (range, 18–71) years: 76 (58%) patients were AML, 28 (21.4%) with ALL, 23 (17.6%) with MDS, and 4 (3.1%) with CML. All patients received calcineurin inhibitors with short-course methotrexate for GVHD prophylaxis. A total of 31 (23.7%) patients received anti-thymocyte globulin. All patients received antibiotics prior to HSCT: 70 (53.4%) patients received glycopeptide, 114 (87.0%) received cefepime, 87 (66.4%) received piperacillin/tazobactam, and 51 (38.9%) received carbapenem. Patients who received glycopeptide had more frequently extensive chronic GVHD (cGVHD) than those who did not (51.1% vs. 28.1% at 5 years) and had more frequently cGVHD of the lung (34.8% vs. 15.8% at 5 years). Pre-transplant use of glycopeptide did not affect the overall survival (OS) or GVHD- and relapse-free survival (GRFS) (median OS; 49 months in glycopeptide group vs. not reached in non-glycopeptide group, p=0.475; median GRFS; 9 months in glycopeptide group vs. 16 months in non-glycopeptide group, p=0.092).

Conclusion: Pre-transplant use of glycopeptide tends to increase the incidence of extensive cGVHD.

KEYWORDS
Glycopeptide; GVHD; hematopoietic stem cell transplantation; hematologic malignancies

Introduction

Allogeneic hematopoietic stem cell transplantation (HSCT) is used to cure several hematological malignancies [1]. However, graft-versus-host disease (GVHD) can cause fatal complications and decrease quality of life and is an obstacle for HSCT. Several factors such as donor type and human leukocyte antigen (HLA) disparity, the intensity of conditioning regimens, and whether immunosuppressants or anti-thymocyte globin (ATG) are used can affect the incidence and extent of acute and chronic GVHD [2, 3].

Many studies have shown that the disruption of fecal microbiota during HSCT influences its outcomes in terms of GVHD [4, 5]. For example, Weber *et al.* showed that decreased urinary excretion of 3-indoxyl sulfate (3-IS), the major conjugate of indole found in humans, reflects a disrupted microbiome [6]. In that study, low 3-IS levels within the first 10 days after allogeneic HSCT were associated with significantly higher transplant-related mortality and worse overall survival. In addition, they mentioned that early initiation of antibiotic treatment was a risk factor for early suppression of 3-IS levels. Shono *et al.* also reported that the use of specific antibiotics for the treatment of febrile neutropenia during allogeneic HSCT, such as imipenem-cilastatin, was associated with GVHD-related mortality and changed the composition of stool microbiota in mouse models [7].

We have used various antibiotics for the treatment of febrile neutropenia during remission induction or consolidation chemotherapy before allogeneic HSCT. The use of antibiotics prior to allogeneic HSCT could change the fecal microbiota environment and affect the outcomes of HSCT. However, no studies have explored this possibility, particularly in terms of GVHD for the patients with hematologic malignancies.

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Methods

Patients

We conducted retrospective analyses of consecutive adult patients with hematological malignancies who received allogeneic HSCT at Chungnam National University Hospital between 2009 and 2018. We excluded patients who received haplo-identical donor or cord blood transplantation. Conditioning regimens were as follows. In the myeloablating conditioning (MAC) regimen, 3.2 mg/kg intravenous (IV) busulfan was administered for 4 days and 40 mg/m^2 fludarabine was administered for 5 days (Bu4Flu) and 3.2 mg/kg IV busulfan was administered for 4 days and 60 mg/kg cyclophosphadmie was administered for 2 days (BuCy). In the reduced intensity conditioning (RIC) regimen, 3.2 mg/kg IV busulfan was administered for 2 days and 30 mg/m^2 fludarabine was administered for 6 days (Bu2Flu). No pharmacokinetic adjustment of busulfan dose was performed. In HLA-matched sibling transplant, cyclosporine for GVHD prophylaxis was given commencing on day −1 with a target level of 150–300 ng/mL. In HLA-matched unrelated donor transplant, tacrolimus was given commencing on day −1 with a target level of 5–15 ng/mL. Short course of methotrexate was given on day +1, +3, +6 and +11 in all patients. Rabbit Anti-thymocyte globulin (thymoglobulin; Sanofi-Aventis, Paris, France) was given from days −3 to −1 at a dose of 1.5 mg/kg in some patients. Most of patients received granulocyte colony-stimulating factor-mobilized peripheral blood stem cells (PBSCs; target CD34+ cell count, 5×10^6/kg). Filgrastim 5ug/kg was administered from day +5 until neutrophil recovery. The study protocol was approved by the Institutional Review Board of Chungnam National University Hospital. The need for informed patient consent was waived given the retrospective nature of the analysis.

Endpoints

Primary endpoints were incidence and severity of acute and chronic GVHD according to the pre-transplant use of antibiotics. Secondary endpoints were overall survival, relapse rate, non-relapse mortality (NRM) and GVHD- and relapse-free survival (GRFS). Acute GVHD was graded according to the modified Seattle Glucksberg criteria and chronic GVHD was graded according to revised Seattle criteria [8, 9]. NRM was defined as death from any cause other than relapse. The composite endpoint of GRFS was defined as extensive chronic GVHD, relapse or death.

Statistics

Categorical variables were compared using chi-square tests and logistic regression was used to evaluate the correlations. Survival was assessed using the Kaplan-Meier method. Survival rates were compared using the log-rank test. Multivariate analyses of independent prognostic factors for survival were performed using the Cox proportional hazard regression model with a 95% confidence interval. A p value < 0.05 was defined as significant. All statistical analyses were performed with the aid of SPSS software ver. 24.0 (IBM Corporation, Armonk, NY, U.S.A.).

Results

Patient characteristics

A total of 131 patients with a median age of 46 years (ranged from 18 to 71) were enrolled. The most common disease in these patients was acute myeloid leukemia (AML), followed by acute lymphoblastic leukemia (ALL) and myelodysplastic syndrome (MDS). About half of the patients received transplantation from sibling donors and the stem cell source was mostly mobilized peripheral blood hematopoietic stem cell. Most of patients (86.3%) received high-resolution DNA matched HLA-identical donor transplantation (eight out of eight alleles or ten out of ten alleles). 115 (87.8%) patients received myeloablating conditioning therapy, including Bu4Flu and BuCy. 31 (23.6%) patients received antithymocyte-globulin (ATG) for the prophylaxis of GVHD. 114 (87.0%), 87 (66.4%), 51 (38.9%) and 70 (53.4%) patients received cefepime, piperacillin/tazobactam, carbapenem and glycopeptide respectively, before allogeneic HSCT for the treatment of febrile neutropenia or infections. The median time from the last use of antibiotics to allogeneic HSCT was 49 days in cefepime, 73 days in piperacillin/tazobactam, 65 days in carbapenem and 60 days in glycopeptide, respectively. The median follow-up duration was 36 months, ranging from 1 to 110 months. Table 1 shows patient demographics.

Pre-transplant antibiotics and acute GVHD

There were no significant correlations between pre-transplant antibiotic usage and the incidence of acute GVHD (Table 2). Patients who received carbapenem and glycopeptide before allogeneic HSCT showed a slightly increased incidence of grade 3–4 acute GVHD compared to those who did not, but the difference was not statistically significant (grade 3–4 acute GVHD, carbapenem vs. no carbapenem, 9.8% vs. 2.5%, p=0.07; glycopeptide vs. not used, 7.1% vs. 3.3%, p=0.327)

Pre-transplant antibiotics and chronic GVHD

Pre-transplant antibiotics did not affect any grade of chronic GVHD (Table 3). However, patients who received glycopeptide before allogeneic HSCT had

Table 1. Baseline characteristics of patients ($N = 131$).

	Patients ($N = 131$)
Age, median (range)	46 (18 – 71) years
Gender, Male: Female (%)	86 (65.6%): 45 (34.4%)
Disease	
AML	76 (58.0%)
MDS	23 (17.6%)
ALL	28 (21.4%)
CML	4 (3.1%)
Disease status at transplant	
1st CR	90 (68.7%)
2nd CR	15 (11.5%)
3rd CR	1 (0.8%)
Refractory	2 (1.5%)
MDS	23 (17.6%)
Donor type	
Sibling	72 (55.0%)
Unrelated	59 (45.0%)
Stem cell source	
Mobilized peripheral blood stem cell	130 (99.2%)
Bone marrow	1 (0.8%)
HLA matching status	
Full matched (8/8 or 10/10)	113 (86.3%)
1 allele mismatched (7/8 or 9/10)	12 (9.2%)
2 allele mismatched (6/8 or 8/10)	6 (4.2%)
Conditioning regimen	
Bu4Flu	112 (85.5%)
Bu2Flu	16 (12.2%)
BuCy	3 (2.3%)
GVHD prophylaxis	
Cyclosporine with MTX	59 (45.0%)
Tacrolimus with MTX	41 (31.3%)
Cyclosporine with MTX plus ATG	13 (9.9%)
Tacrolimus with MTX plus ATG	18 (13.7%)
Pre-transplant use of antibiotics	
Cefepime	114 (87.0%)
Piperacillin/tazobactam	87 (66.4%)
Carbapenem	51 (38.9%)
Glycopeptide	70 (53.4%)
Time from the use of antibiotics to transplantation, median (range)	
Cefepime	49 (5-134) days
Piperacillin/tazobactam	73 (26-147) days
Carbapenem	65 (22-150) days
Glycopeptide	60 (24-125) days
HCT-CI, median (range)	0 (0 – 3)
F/U duration, median (range)	36 (1 - 110) months

more frequent extensive chronic GVHD than those who did not. The 5-year cumulative incidence of extensive chronic GVHD was 51.1% in the glycopeptide group, significantly higher than the 28.1% in the non-glycopeptide group (p=0.026, Figure 1). Interestingly, patients who received glycopeptide had more frequent chronic lung GVHD than those who did not. The 5-year cumulative incidence of chronic lung GVHD was 34.8% in the glycopeptide group and 15.8% in the non-glycopeptide group, with a p value of 0.028 (Figure 2). Adverse risk factors for extensive chronic GVHD were the pre-transplant use of glycopeptide, HLA-mismatched donor, non-use of ATG, and grade 3–4 acute GVHD based on univariate analyses. However, in multivariate analyses, the poor prognostic factors that affected extensive chronic GVHD were mismatched donor, non-use of ATG, and severe acute GVHD (hazard ratio 2.2 [95% CI, 1.12–4.35], 3.73 [95% CI, 1.33–10.47], and 2.64 [95% CI, 1.66–4.19], respectively, Table 4). And there was no significant risk factor in the multivariate analyses for chronic lung GVHD (Table 5).

Table 2. Correlation between pre-transplant antiobiotics and acute GVHD.

Antibiotics	Acute GVHD, all grade	p value†	Acute GVHD, grade 3–4	p value†
Cefepime *vs.* None	31.6% *vs.* 23.5%	0.501	5.3% *vs.* 5.9%	0.916
Piperacillin/ tazobactam *vs.* None	28.7% *vs.* 34.1%	0.530	5.7% *vs.* 4.5%	0.773
Carbapenem *vs.* None	33.3% *vs.* 28.8%	0.579	9.8% *vs.* 2.5%	0.070
Glycopeptide *vs.* None	24.3% *vs.* 37.7%	0.096	7.1% *vs.* 3.3%	0.327

†, χ^2 test.

Table 3. Correlation between pre-transplant antiobiotics and chronic GVHD.

Antibiotics	Chronic GVHD, any grade (limited + extensive)	p value†
Cefepime *vs.* None	70.2% *vs.* 76.5%	0.594
Piperacillin/tazobactam *vs.* None	70.1% *vs.* 72.7%	0.756
Carbapenem *vs.* None	68.6% *vs.* 72.5%	0.634
Glycopeptide *vs.* None	70.5% *vs.* 71.4%	0.906

†, χ^2 test.

Pre-transplant use of glycopeptide and survival

Patients who received glycopeptide before allogeneic HSCT had slightly decreased overall survival compared to those who did not, but the difference was not statistically significant (median OS; 49 months in glycopeptide group vs. not reached in non-glycopeptide group, $p = 0.475$, Figure 3A). The cumulative incidence of relapse did not differ between the two groups (cumulative incidence of relapse at 5-yr; 35.2% in glycopeptide group vs. 29.3% in non-glycopeptide group, $p = 0.868$, Figure 3B). Non-relapse mortality was not changed by pre-transplant use of glycopeptide (cumulative incidence of NRM at 5-yr; 19.6% in glycopeptide group vs. 18.8% in non-glycopeptide group, $p = 0.973$, Figure 3C). Patients who received glycopeptide before allogeneic HSCT had slightly decreased GVHD- and relapse-free survival compared to those who did not, but the difference was not statistically significant (median GRFS; 9 months in glycopeptide group vs. 16 months in non-glycopeptide group, $p = 0.092$, Figure 3D).

Discussion

We found that pre-transplant use of glycopeptide may be correlated with extensive chronic GVHD, particularly in the lung. In our study, those who used glycopeptide before allogeneic HSCT had an approximately two-fold higher 5-year cumulative incidence of extensive chronic GVHD than those who did not. This may be related to a disruption of gut microbiota. Microbiota play an important role in immune homeostasis, and disruption of the gut microbiota

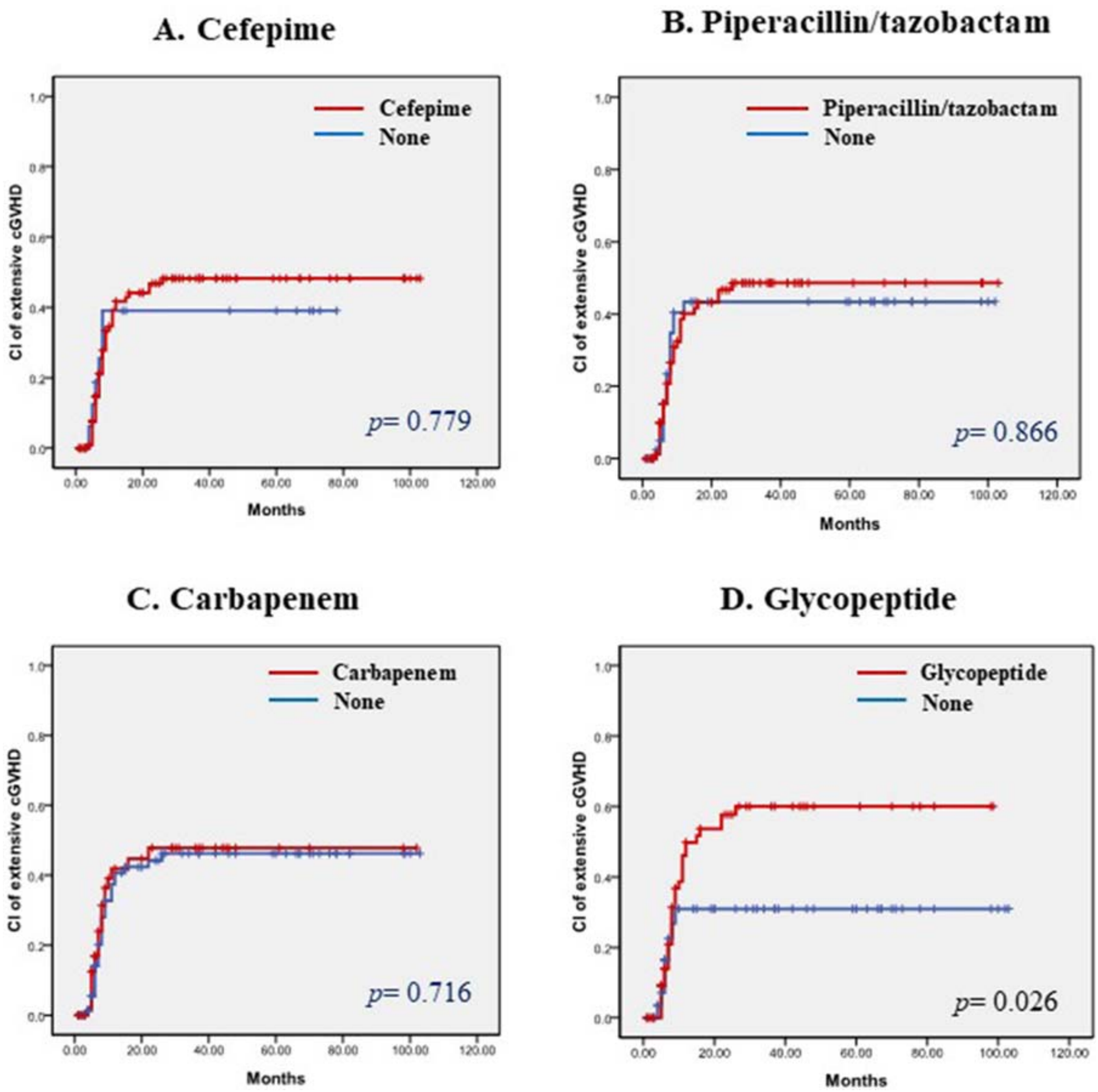

Figure 1. (A-D). Cumulative incidence of extensive chronic GVHD according to pre-transplant use of antibiotics. (A) Cefepime, (B) Piperacillin/tazobactam, (C) Carbapenem, and (D) Glycopeptide.

may affect immune reconstitution after allogeneic HSCT and could be correlated with acute and chronic GVHD [10, 11]. For example, Chen *et al*. reported that changes in the intestinal microbiota composition are associated with acute GVHD [12, 13]. In addition, Holler *et al*. showed that increases in the proportion of *Enterococcus* and decreases in commensal bacteria are common in patients who develop gastrointestinal GVHD [14]. And, Jenq *et al*. reported that increased bacterial diversity is associated with reduced GVHD-related mortality [15].

Most of patients with hematological malignancies commonly underwent broad-spectrum antibiotic therapy before allogeneic HSCT for the treatment of febrile neutropenia. Indeed, most patients were exposed to antibiotics before HSCT in our study, and these antibiotics might have affected the fecal microbiota [16, 17]. Broad-spectrum antibiotics could not only reduce bacterial diversity but also increase resistant bacteria, and enable intrusion of pathogenic organisms through the depletion of natural intestinal microbiota. For example, Ferrer *et al*. showed that specific bacterial lineages, such as *Enterococcus*, *Blautia*, *Faecalibacterium*, and *Akkermansia*, are present in patients who receive β-lactam antibiotic intervention [18]. And, Scott *et al*. showed that disruption of the gut microbiota by antibiotics reduces butyrate, which has protective effects against antibiotic-associated immune dysfunction in mice, in the intestine [17, 19].

Fecal microbiota in patients with hematologic malignancies before allogeneic HSCT differs from that of the normal population. In the report of Kusakabe *et al*., populations of butyrate-producing bacteria were significantly lower in the fecal samples of patients before allogenic HSCT than in healthy controls. In addition, the population of *Enterococcus* was significantly higher in pre-transplant gut

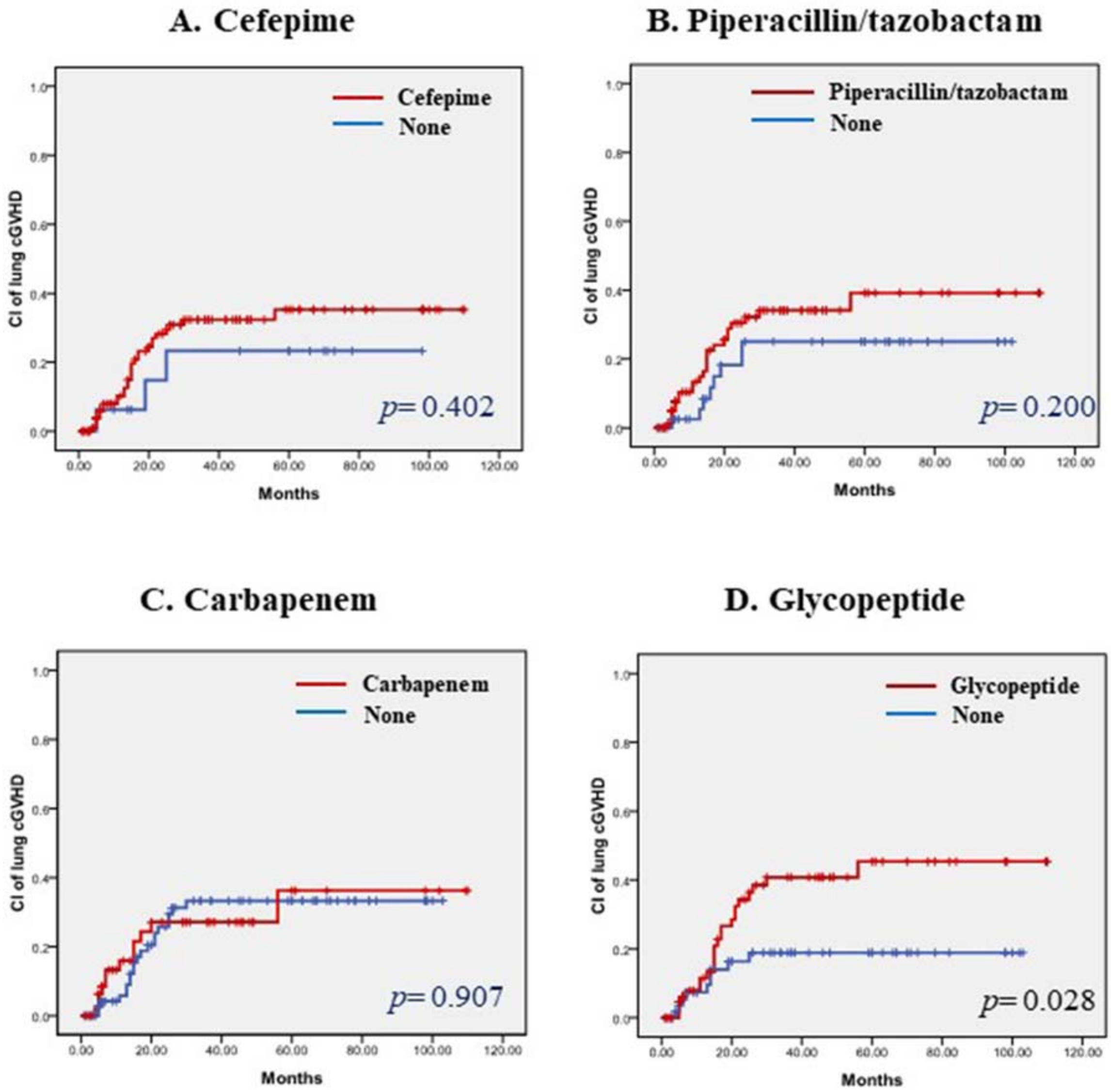

Figure 2. (A-D). Cumulative incidence of chronic lung GVHD according to pre-transplant use of antibiotics. (A) Cefepime, (B) Piperacillin/tazobactam, (C) Carbapenem, and (D) Glycopeptide.

Table 4. Poor risk factors for extensive chronic GVHD in univariate and multivariate analysis.

Parameters	Incidence of extensive cGHVD	Univariate ($p^{§}$)	Multivariate (p*)	HR* (95% CI)
Pre-transplant use of glycopeptide *vs* non-use	51.1% *vs* 28.1%	0.026	0.157	1.55 (0.84-2.84)
Mismatched donor *vs* matched	61.1% *vs* 38.4%	0.035	0.022	2.20 (1.12-4.35)
Non-use of ATG *vs* use	51.0% *vs* 12.9%	0.003	0.012	3.73 (1.33-10.47)
Grade 3–4 of Acute GVHD *vs* grade 0–2	100% *vs* 43.2%	< 0.001	< 0.001	2.64 (1.66-4.19)
Unrelated donor *vs* sibling	42.4% *vs* 34.7%	0.554	–	–
Myeloablating conditioning *vs* reduced intensity	40.9% *vs* 18.7%	0.284	–	–

§Log Rank test, *Cox proportional hazard ratio.

microbiotas [20]. And, Peled *et al*. demonstrated that the diversity of intestinal microbiota before transplantation was quite variable, and a higher diversity at the time of neutrophil engraftment was associated with lower mortality [21]. Glycopeptide is mainly used for the treatment of methicillin-resistant

Table 5. Poor risk factors for chronic lung GVHD in univariate and multivariate analysis.

Parameters	Incidence of lung GHVD	Univariate ($p^{§}$)	Multivariate (p*)	HR* (95% CI)
Pre-transplant use of glycopeptide *vs* non-use	34.8% *vs* 15.8%	0.028	0.062	1.45 (0.98-2.15)
Mismatched donor *vs* matched	50.0% *vs* 24.7%	0.089	0.084	2.05 (0.90-4.64)
Non-use of ATG *vs* use	32.9% *vs* 12.5%	0.075	0.146	2.44 (0.73-8.17)
Grade 3–4 of Acute GVHD *vs* grade 0–2	50.0% *vs* 27.2%	0.079	0.146	2.46 (0.73-8.33)
Unrelated donor *vs* sibling	30.8% *vs* 26.3%	0.721	–	–
Myeloablating conditioning *vs* reduced intensity	28.3% *vs* 30.0%	0.897	–	–

§Log Rank test, *Cox proportional hazard ratio.

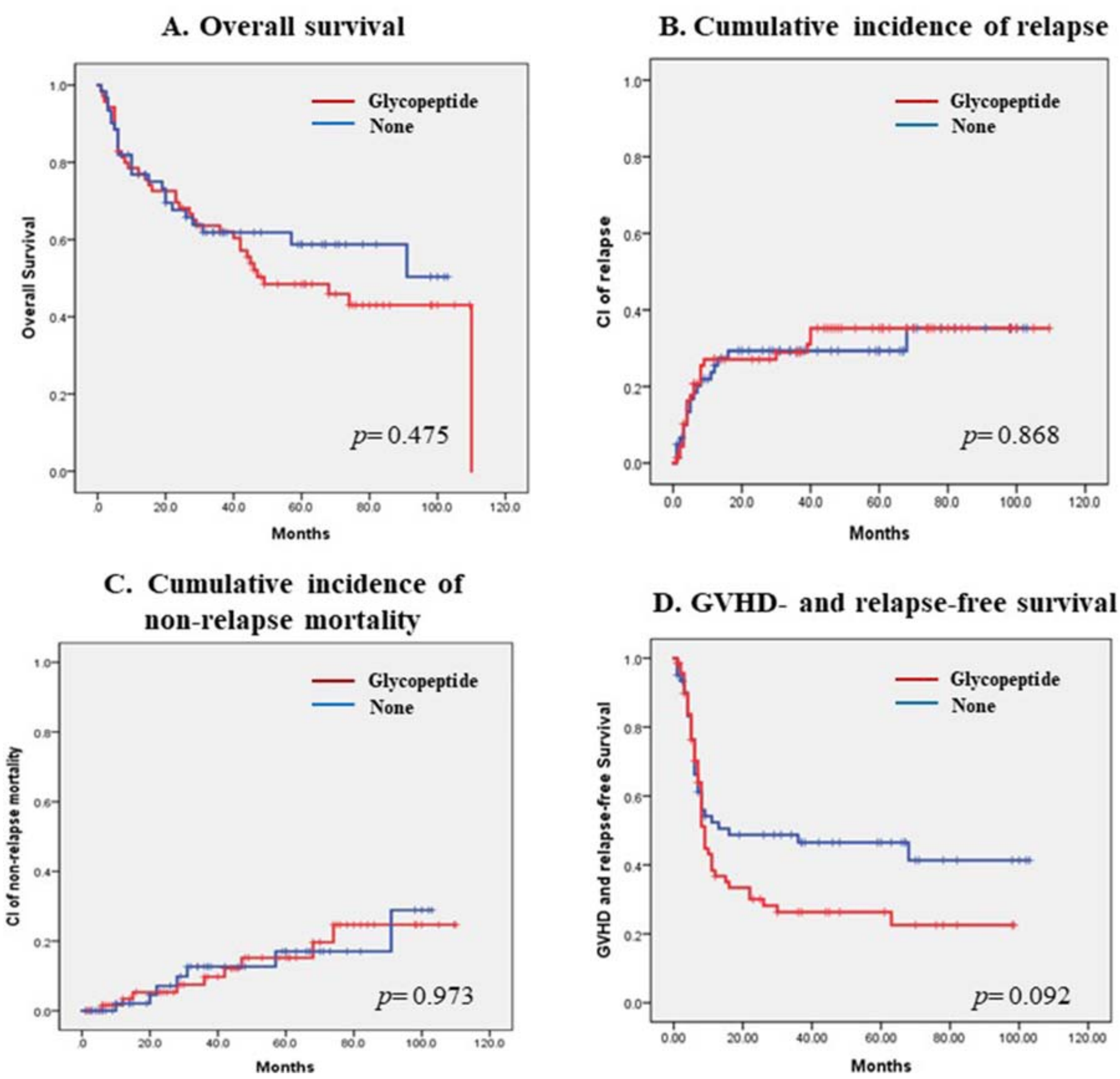

Figure 3. (A-D). (A) Overall survival, (B) Cumulative incidence of relapse, (C) Cumulative incidence of non-relapse mortality and (D) GVHD- and relapse-free survival according to pre-transplant use of glycopeptide.

Staphylococcus aureus and pneumonia in febrile neutropenia during chemotherapy according to IDSA guidelines [22, 23]. About half of our patients were treated with glycopeptide before allogeneic HSCT. This high usage rate of glycopeptide may have been due to the high rate of acute leukemia in our study population. Patients who had acute leukemia mostly experienced febrile neutropenia during the induction and consolidation chemotherapy. In this study, we showed the correlation between glycopeptide and inflammation of lung through the result that patients treated with glycopeptide had significantly increased extensive chronic GVHD of the lung. Some studies also have shown that vancomycin-induced gut dysbiosis during *Pseudomonas aeruginosa* pulmonary infection influences the lung-gut immunologic axis [24]. However, this study did not reveal a specific mechanism, so further research is needed in the future for investigating the exact correlation between antibiotics and lung inflammation in patients underwent allogenic HSCT.

Pre-transplant use of glycopeptide increased the incidence of extensive chronic GVHD in univariate analyses in our study; however, the statistical power was weakened in multivariate analyses. This suggests that future studies with a larger number of patients are required. And a high-throughput molecular test for examining the composition of fecal microbiome is additionally needed to investigate whether pre-transplant use of glycopeptide actually changes the fecal microbiota.

In conclusion, this study shows the tendancy between pre-transplant antibiotic use and the incidence and severity of chronic GVHD in patients who underwent allogenic HSCT. We suggest that patients who are treated with glycopeptide prior to allogenic HSCT should be carefully monitored for the occurrence of extensive chronic GVHD.

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Disclosure statement

No potential conflict of interest was reported by the author (s).

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